



ME 603: Advanced Fluid Mechanics

Course Code	ME 603
Course Name	Advanced Fluid Mechanics
L-T-P-C	3-0-0-3
Prerequisites	Fluid Mechanics
Intended for	MS/PhD
Distribution	Elective
Approval	2nd Adhoc Senate (May 2012)

Course Contents

- **Introduction**

Eulerian and Lagrangian description of flow; Motion of fluid element translation, rotation and deformation; vorticity and strain-rate tensors; Continuity equation, Cauchy's equations of motion, Derivation of Navier-Stokes equations for compressible flow. [Lectures]

- **Exact solutions of Navier-Stokes equations**

Plane Poiseuille flow and Couette flow, Hagen-Poiseuille flow, flow between two concentric rotating cylinders, Stokes first and second problems, flow near a rotating disk, flow in convergent-divergent channels. [Lectures]

- **Slow viscous flow**

Stokes and Oseen's approximation, theory of hydrodynamic lubrication. [Lectures]

- **Boundary layer Analysis**

Derivation of boundary layer equations; Exact solutions; Approximate methods; Momentum integral method. [Lectures]

- **Stability**

Introduction to hydrodynamic stability, Orr-Sommerfeld equation. [Lectures]

- **Introduction to Turbulence**

Description of turbulent flow, averaging, RANS, Introduction to turbulent models, Empirical laws. [Lectures]

Text and Reference Books

1. White F M, **Viscous Fluid Flow**, 3rd Edition, Tata McGrawhill, 2011.
2. Cebeci T and Bradshaw P, **Momentum Transfer in Boundary Layers**, McGrawHill, 1977.
3. Schlichting H and Gersten K, **Boundary Layer theory**, 8th Edition, Springer, 2000.
4. Kundu P K and Cohen I M, **Fluid Mechanics**, 4th Edition, Elsevier, 2005.